



Problem solving with numerical summaries

Task A: Critically evaluating statistical conclusions

- 1) The table shows a statistical summary of data from The Met Office from 1941-2022. The data shows the amount of rainfall (mm) per month in Durham and Lerwick.

Make two comparisons between the amounts of rainfall in Durham and Lerwick.

monthly rainfall (mm)	Durham	Lerwick
mean	53.6	98.3
median	48.4	92
range	194.2	301.7

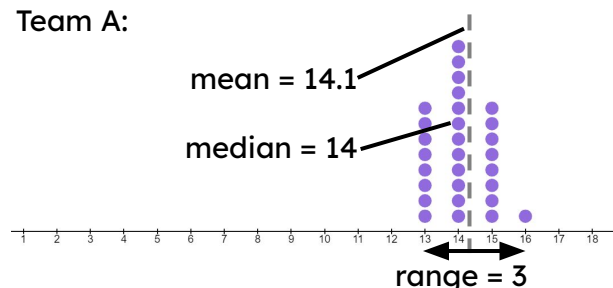
The mean for Lerwick (98.3) is greater than the mean for Durham (53.6), meaning that Lerwick got more rainfall on average.

The range for Durham (194.2) is less than the range for Lerwick (301.7), meaning that Durham's rainfall is more consistent/predictable.

Alternative option: the median could be used instead of the mean.

- 2) Two classes practised throwing basketballs into a hoop. Each person had 20 attempts. The dot plots show the number of points each person scored. Lucas and Aisha used the data to make conclusions.

Team A:



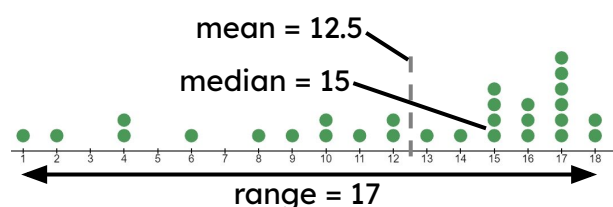
Lucas

Team B scored higher than Team A, on average.

Aisha

Team A's scores were more consistent.

Team B:



- a) Give evidence to support Lucas's conclusion.

The median for Team B (15) is greater than the median for Team A (14).

- b) Give evidence to counter Lucas's conclusion.

The mean for Team B (12.5) is less than the mean for Team A (14.1).

- c) Give evidence to support Aisha's conclusion.

The range for Team A (3) is less than the range for Team B (17).



Task B: Considering context behind statistical conclusions

- 1) Alex has represented his school and his county in swimming races.



When I raced for my school, I got a gold medal.
But then when I raced for my county, I only got a bronze medal.
That must mean I swam slower when racing for my county.

Explain why Alex's conclusion could be wrong.

Alex may have still swum at the same speed but against faster opponents when racing for their county than school.

- 2) Laura is looking at her school reading book records.



When I was 6 years old, I read an average of 9.2 books per month.
I now read an average of 1.4 books per month.
I must be a slower reader now than I was when I was 6 years old.

Explain why Laura's conclusion could be wrong.

Books are different lengths. The books Laura read when she was 6 years old were probably shorter than the books she is reading now.

- 3) A survey found that 18% of workers in the town of Scarborough walk to work and that 7% of workers in the City of London walk to work.



There are more people walking to work in Scarborough than in London.

Izzy

Explain why Izzy's conclusion could be wrong.

*The populations of cities are usually greater than the population of towns.
7% of a large population could be more than 18% of a small population.*

- 4) A survey in 1971 found that 15% of workers in Liverpool travelled to work by bus.
A survey in 2023 found that 11% of workers in Liverpool travelled to work by bus.



Liverpool does not need to run as many buses as they did in 1971
because there are fewer people using them to travel to work.

Jacob

Explain why Jacob's conclusion could be wrong.

*The population of Liverpool may have increased since 1971.
11% of a large population could be more than 15% of a small population.*

